FINANCIAL MANAGEMENT

Suggested Answers
July-August 2022

Answer to the Question# 1 (a):

Financial Analysis whether to set up the manufacturing units in Bangladesh or not may be carried using NPV technique as follows:

Determination of NPV

Year	CFAT ($ Million)	PVF@12%	PV($ Million)
1-5	408.75(W-4)	3.6048	1,473.46
5	52.5 (W-2)	0.5674	29.79
			1,503.25
Less: Initial Outflow (W-1)			802.5
			103.0822

Since NPV is positive the proposal should be accepted.

Working- 1: Incremental Cash Outflows

	$ Million
Cost of Plant and Machinery	750.00
Working Capital	75.00
Release of existing Working Capital	(22.50)
	802.5

Working – 2: Incremental Cash Inflow after Tax (CFAT)

(a) Generated by investment in Bangladesh for 5 years

	$ Million
Sales Revenue (7.5 Million x $120)	900.00
Less: Costs	
Variable Cost (7.5 Million x $30)	225.00
Fixed Cost	45.00
Depreciation ($750 Million/5)	150.00
EBIT	480.00
Taxes@25%	120.00
EAT	360.00
Add: Depreciation	150.00
CFAT (1-5 years)	510.00
Cash flow at the end of the 5 years (Release of Working Capital)	52.5

Working- 3: Cash generation by exports if no subsidiary is established

	$ Million
Sales Revenue (2.25 Million x $ 120)	270.00
Less: Variable Cost (2.25 Million x $60)	135.00
Contribution before tax	135.00
Tax@25%	33.75
CFAT (1-5 years)	101.25

Working- 4: Additional CFAT attributable to Foreign Investment

	$ Million
Through setting up subsidiary in Bangladesh (W-2)	510.00
Through Exports in Bangladesh (W-3)	101.25
CFAT (1-5 years)	408.75

Answer to the Question# 1 (b) (i):

Selection of Projects

Project		Fund Requirement	Investment	Rem. Funds
BA	-	84	84	276
AC	-	60	60	216
AB	-	96	96	120
BC	-	90	90	30
AA	-	84	30	
			360	

Answer to the Question# 1 (b) (ii):

If projects are indivisible, it must be done in its entirety or not at all and for the optimum use of capital investment**,** a trial-and-error method could be and the ranking of projects under capital rationing is more difficult than ranking divisible projects.

In such a situation depending on the available funds the best combination of projects i.e. the best portfolio which will maximize the NPV should be considered. Any remaining funds should be invested in financial securities rather than allowing them to idle in order to enhance earnings or profitability of the available funds.

Answer to the Question# 1 (c) (i):

To find the value of the target to the acquirer, we need to find the share price with the new growth rate. We begin by finding the required return for shareholders of the target firm.
The earnings per share of the target are:

EPSP = Tk. 960,000/750,000 = Tk. 1.28 per share

The price per share is:

PP = 10(Tk. 1.28) = Tk. 12.80

And the dividends per share are:

DPSP = Tk. 470,000/750,000 = Tk. 63

The current required return for Palmer shareholders, which incorporates the risk of the company is:

RE = (Tk. 63(1.04)/ Tk. 12.80) + .04 = .0909

The price per share of Palmer with the new growth rate is:

PP = Tk. .73(1.06)/ (.0909 - .06) = Tk. 21.49

The value of the target firm to the acquiring firm is the number of shares outstanding times the price per share under the new growth rate assumptions, so:

V0T = 750,000 (Tk. 21.49) = Tk. 16,114,285.71

Answer to the Question# 1 (c) (ii):

The gain to the acquiring firm will be the value of the target firm to the acquiring firm minus the market value of the target, so:

Gain = Tk. 16,114,285.71 – 750,000 (Tk.12.80) = Tk. 6,51,285.71

Answer to the Question# 1 (c) (iii):

The NPV of the acquisition is the value of the target firm to the acquiring firm minus the cost of the acquisition, is:

NPV = Tk. 16,114,285.71 - 750,000 (Tk. 20) = Tk. 1,114,285.71

Answer to the Question# 1 (c) (iv):

The most the acquiring firm should be willing to pay per share is the offer price per share plus the NPV per share, so:

Maximum bid price = Tk.20 + (Tk.1,114,285.71/750,000) = Tk. 21.49

Notice that this is the same value we calculated earlier in part a as the value of the target to the acquirer.

Answer to the Question# 1 (c) (v):

The price of the stock in the merged firm would be the market value of the acquiring firm plus the value of the target to the acquirer, divided by the number of shares in the merged firm, so:

PFP = (Tk. 60,900,000 + 16,114,285.71)/ (1,500,000 + 225,000) = Tk.44.65

The NPV of the stock offer is the value of the target to the acquirer minus the value offered to the target shareholders. The value offered to the target shareholders is the stock price of the merged firm times the number of shares offered, so:

NPV = Tk.16,114,285.71 - 225,000(Tk.44.65) = Tk.6,068,944.10

Answer to the Question# 1 (c) (vi):

Yes, the acquisition should go forward, and Plant should offer share is since the NPV is higher.

Answer to the Question# 1 (c) (vii):

Using the new growth rate in the dividend growth model, along with the dividend and required return we calculated earlier, the price of the target under these assumptions is:

PP = Tk. .63(1.05)/ (.0909 – .05) – Tk. 16.08

And the value of the target firm to the acquiring firm is:

VP* = 750,000 (Tk. 6.08) = Tk.12,061,099.80

The gain to the acquiring firm will be:

Gain = Tk.12,061 ,099.80 - 750,000 (Tk. 12.80) =Tk.2,461,099.80

The NPV of the cash offer is now:

NPV cash = Tk. 12,06 1,099.80 – 750,000 (Tk. 20) = Tk. 2,938,900.20

And the new price per share of the merged firm will be:

PFP = (Tk. 60,900,000 + 12,061,099.80) / (1,500,000 + 225,000) = Tk. 42.30

And the NPV of the stock offer under the new assumption will be:

NPV stock = Tk. 12,061,09980 – 225,000 (Tk. 42.30) = Tk.2,244,434.61

Even with the lower projected growth rate, the stock offer still has a positive NPV. The NPV of the stock offer is still higher. Plant should purchase Palmer with a stock offer of 225,000 shares.

Answer to the Question# 2 (a) (i):

<u>Option 1 – Through foreign exchange market:</u>

2 month forward contract with the banker, to sell ¥ 30,000,000 proceeds from Japan

With the forward contract, the company can sell ¥ 30,000,000 at the bank's buying rate of ¥ 1.3755 per BDT.

BDT proceeds under forward contract = ¥30,000,000 * 1.3755

= BDT 41,265,000

Rupee proceeds under future spot rate (bankers buying rate):

Current Spot Rate BDT/¥ = 1.4185

In two months time, the expected spot rate = 1.4185 * 0 .94 = 1.3334

BDT proceeds at the spot rate in 2 moths time = ¥ 30,000,000 * 1.3334

= BDT 40,002,000

Expected gain from forward contract = BDT 41,265,000 - BDT 40,002,000

= BDT 1,263,000

Option 2 – Through money market hedging

The company can borrow ¥ 30,000,000 at the rate of 7.5% for two months and convert it into Taka at today's spot rate and then invest the Taka proceeds at the rate of 12% for two months. Finally, the company can settle the yen loan from the yen proceeds in two months' time.

Amount that could be borrowed in yen = ¥30,000,000/(1.0125)
= ¥ 29,629,630

BDT proceeds from conversion = ¥29,629,630 * (1.4185)
= BDT 42,029,630

Investing the Taka proceeds in the money market = BDT 42,029,630 * (1.02)
= BDT 42,870,223

Gain from money market hedging = BDT 42,870,223 - BDT 40,002,000

= BDT 2,868,223

The company should go for money market hedging as the gain from this option is greater than the gain arising from using a forward contract.

Answer to the Question# 2 (a) (ii):

Three different forms of currency risks a firm could face are: transaction risk, translation risk and economic risk.

Transaction Risk: This is the risk of making losses due to exchange rate changes in the course of international transactions. It occurs when prices are agreed in foreign currency terms in advance and the payment or the settlement is made at a later date and adverse changes in the exchange rate takes place between the two dates.
Translation Risk: This is the risk of exchange losses that could take place when the accounting figures of foreign operations (may be branches) are translated into the domestic currency in the preparation of financial reports (restating profit/loss and assets/liabilities of foreign branches at the exchange rate on the reporting date of financial reports).

Economic Risk: Change in the exchange rate will have an impact on most firms' competitiveness in numerous ways, to the extent their cash flows and their input/output are exposed to foreign currency fluctuations e.g. agro businesses will be subject to such economic risk due to their input (fertilizer and agro chemical) prices increasing as a result of exchange rate depreciations. On the other hand, if the rupee appreciates against foreign currencies, the tourism industry will suffer an economic risk due to decreased tourist arrivals**.**

Answer to the Question# 2 (b) (i):

The profit on the tender (if J Limited is successful in winning) is calculated as follows:

Details	Amount	Forward Rate	€ Value
Receipt 1	300,000,000 ZAR (40,000*750)	12.56 (13.00 - .44)	2,388,535
Receipt 2	450,000,000 ZAR (60,000*750)	12.1 (13.00 - .9)	3,719,008
Payment	10,000,000 AD (100,000*100)	2.08 (1.9 + .18)	-4,807,692
Profit On Contract			**1,299,851**

Answer to the Question# 2 (b) (ii):

Currency & Exercise Dates	Amount of Foreign Currency	Exercise Price	Options Purchased	€ Value	Commission	Final € Receipt/Cost
6 Month ZAR Put	300,000,000 ZAR	12	60	2,500,000	-75,000	2,425,000
12 Month ZAR Put	450,000,000 ZAR	12	90	3,750,000	-247,500	3,502,500
12 Month AD Call	10,000,000 AD	2	100	-5,000,000	-80,000	-5,080,000
Profit On Tender						**847,000**

Answer to the Question# 3 (a):

Continuing to obtain cover from its own Denver Cover Plant would allow Gaint Auto to maintain its current level of control over the quality of the covers and the timing of their delivery. Keeping the Denver Cover Plant open also allows Gaint Auto more flexibility than purchasing the coverings from outside suppliers. Gant Auto could more easily alter the coverings design and change the quantities produced, especially if long-term contracts are required with outside suppliers. Gaint Auto should also consider the economic impact that closing Denver Cover with have on the community and how this might affect Gaint Auto's other operations in the region.

Answer to the Question# 3 (b) (i):

The following costs can be avoided by closing the plant, and therefore are relevant to the decision:

Materials		Tk. 14,000,000
Labor:		
Direct	Tk. 13,100,000	
Supervision	900,000	
Indirect plant	4,000,000	18,000,000
Differential pension cost		20,000,000
(Tk. 5,000,000 – 3,000,000)		2,000,000
Total annual relevant costs		Tk. 34,000,000

Answer to the Question# 3 (b) (ii):

The following costs can be avoided by closing the plant, and therefore are not relevant to the decision:

Depreciation – equipment	Tk. 3,200,000
Depreciation – building	7,000,000
Continuing pension cost (Tk.5,000,000–2,000,000)	3,000,000
Plant manager and staff	800,000
Corporate allocation	4,000,000
Total annual continuing costs	Tk. 18,000,000

The depreciation amounts are not relevant to the decision because they are sunk costs. Three-fifths of the annual pension expense (Tk.3,000,000) is not relevant because it would continue whether or not the plant is closed. The amount for plant manager and staff is not relevant because Rahath and his staff would continue with Gant Auto and administer the three remaining plants. The corporate allocation is not relevant because this represents costs incurred outside Denver Cover and assigned to the plant.

Answer to the Question# 3 (b) (iii):

The following nonrecurring costs would arise in the year that the plant is closed, but would not be incurred in any other year:

Termination charges on canceled material orders	
(Tk. 14,000,000×20%	Tk. 2,800,000
Employment assistance	1,500,000
Total recurring costs	Tk. 4,300,000

These two costs are relevant to the decision because they will be incurred only if the plant is closed.

Answer to the Question# 3 (c):

No. the plant should not be closed. The computations are:

Cost of purchasing the coves outside	Tk. (35,000,000)	Tk. (35,000,000)
Costs avoided by closing the pant (Part 2a)	34,000,000	34,000,000
Cost of closing the plant (first year only)	(4,300,000	-
Net advantage (disadvantage) of closing the plant	Tk. (5,300,000)	Tk. (1,000,000)

Answer to the Question# 3 (d):

Items not specifically mentioned in the text that should be considered by Giant Auto before making a decision include:

- The disposal value or alternate uses of the building and equipment.
- Any income tax implications
- The outside supplier's prices in future years.
- The cost to manufacture coverings at the Denver Cover Plant in future years.
- The value of the time Rahath and his staff would have spent managing the Denver Cover Plant. This time may be spent on other important matters.

Answer to the Question# 4 (a):

The effects of the proposal should be to increase income by: (1) Increasing profits from sales, (2) reducing the opportunity cost of investing in debtors and (3) reducing the cost of bad debts. We will analyze each of each of these in turn.

(1) Incremental profit contribution

Increase in sales	= Tk15m × 0.05 =	750,000
Increase in variable costs	= Tk0.75m × 0.60 =	450,000
Increase in profit contribution	=	300,000

(2) Investment in debtors Average existing investment

Tk15m × (30/365)	= 1,232,877	
Average proposed investment	= 863.014	
(Tk 15m × 1.05) × (20/365)		
Reduction	369,863	
Cost saving in investment	= 369,863×0.15=58,379	

3) Bad debts

Existing cost: Tk15m × 0.02	= 300,000	
Proposal cost: (Tk15m×1.05)×0.015	= 238,250	
Saving		
		= 63,750
Total Incremental Income		419,229

*The average collection period under existing arrangements is 30 days.

Incremental costs

Costs will increase due to the cost of the cash discount itself.

Cost of cash discount	
[(Tk15m × 1.05) × 0.70] × 0.02	= 220,500
Incremental Profit	198,729

As the proposal is expected to yield a net annual increase of almost Tk200,000 in the company profits (ignoring tax) it will add to the value of the company and should therefore be account. The value of the proposal can be determined using NPV principles.

NPV approach

The NPV of the proposal can be calculated by comparing incremental cash inflows with incremental cash outflows. In this example they would be

Annual cash inflows	Taka
Incremental profit contribution	300,000
Bad debt reduction	63,750
	363,750
Annual cash outflows	Taka
Cash discount	220,500
Net cash flow	143,250

The proposal is expected to yield an annual increase in pre-tax cash flows of Tk143,250 approximate value of this cash flow to the company can be calculated as a perpetuity using per cent cost of capital and assuming a tax rate of 30 per cent.

$$\frac{\int 143{,}250 \times (1-0.30)}{0.15} = \int 668{,}500$$

Remember it is only the after-tax cash flows which are relevant to the company. In addition, there is the one-off benefit of the reduction in the average amount of cash tied up in debts. This will provide the company with a non-recurring initial cash inflow of Tk369,863, therefore the total value of the proposal is Tk1,038,363 (Tk668,500+Tk369,863). The market value of the company would be expected to increase by this amount. As the project is clearly wealth enhancing it should be implemented.

Answer to the Question# 4 (b) (i):

Set up a timeline like the one in the preceding problem.

1/1/x5	1/1/x6	1/1/x7	1/1/x8	1/1/x9
	PV = ?			1,000

Note that your deposit will grow for three years at 8 percent. The fact that it is now January 1, 20x5, is irrelevant. The deposit on January 1, 20x6, is the PV, and the FV is Tk. 1,000. Here is the solution:

N = 3; I = 8%; PMT = 0; FV = Tk. 1,000; PV=?; PV=Tk. 793.83

$$\text{PV= Tk. } 1{,}000\left[\frac{1}{(1.08)^3}\right] = \text{Tk.}1{,}000(0.79383) = \text{Tk.}793.83$$

= Initial deposit to accumulate Tk. 1,000.

Answer to the Question# 4 (b) (ii):

1/1/x5	1/1/x6	1/1/x7	1/1/x8	1/1/x9
	PMT	PMT	PMT	PMT
				FV = 1,000

Here we are dealing with a four-year annuity whose first payment occurs one year from today, on 1/1/x6, and whose future value must equal Tk. 1,000. You should modify the timeline to help visualize the situation. Here is the solution:

N = 4; I = 8%; PV = 0; FV = Tk. 1,000; PMT=? PMT=Tk. 221.92

$$FVA_4 = PMT\left[\frac{(1.08)^4 - 1}{(1.08)^3}\right] = Tk.1{,}000$$

$$PMT = \frac{Tk.1{,}000}{4.50611}$$

$$= \frac{Tk.1{,}000}{4.50611} = Tk.221.92$$

= Payment necessary to accumulate Tk. 1,000.

Answer to the Question# 4 (b) (iii):

This problem can be approached in several ways. Perhaps the simplest is to ask this question: "If I received Tk. 750 on 1/1/x6 and deposited it to earn 8 percent, would I have acquired Tk. 1,000 on 1/1/x9?" The answer is no:

1/1/x5	8%	1/1/x6	1/1/x7	1/1/x8	1/1/x9
		-750			FV= ?

FV3 = Tk. 750 (1.08)3 = Tk. 944.78.

This indicates that you should let your father make the payments rather than accept the lump sum of Tk. 750. You could also compare the Tk. 750 with the PV of the payments:

PV=Tk.735.03

1/1/x5	8%	1/1/x6	1/1/x7	1/1/x8	1/1/x9
PV = ?		221.92	221.92	221.92	221.92

N=4; I=8%; PMT = - Tk 221.92; FV = 0, PV =? PV = TK 735.03

$$PVA4 = Tk.\ 735.03 = Tk.\ 221.92\left[\frac{1 - \frac{1}{(1.08)^4}}{0.08}\right]$$

Tk. 221.92 (3.31212) = Tk. 735.03 = Present value at 1/1/01 of the required payments.

This is less than the Tk. 750 lump sum offer, so your initial reaction might be to accept the lump sum of Tk. 750. However, this would be a mistake. The problem is that when you found the Tk. 735.03 PV of the annuity, you were finding the value of the annuity today, on January 1, 2005. You were comparing Tk. 735.03 today with the lump sum of Tk. 750 one year from now. This is, of course, invalid. What you should have done was take the Tk. 735.03, recognize that this is the PV of an annuity as of January 1, 2005, multiply Tk. 735.03 by 1.08 to get Tk. 93.83, and compare Tk. 793.83 with the lump sum of Tk. 750. You would then take your father's offer to make the payments rather than take the lump sum on January 1, 2006. If you solved the PV for an annuity due, you would find the same answer.

Answer to the Question# 4 (b) (iv):

1/1/x5	k=?	1/1/x6	1/1/x7	1/1/x8	1/1/x9
		-750			1000

N = 3; PV = -750; PMT = 0; FV = Tk. 1,000; I = ?; I = 10.0642%

Answer to the Question# 4 (b) (v):

1/1/x5	k=?	1/1/x6	1/1/x7	1/1/x8	1/1/x9
		186.29	186.29	186.29	186.29
					FV =1,000

N = 4; PV = 0; PMT = -Tk. 186.29: FV = Tk. 1,000;
I = ?; I = 19.9997%.

It might be able to find a borrower willing to offer you a 20 percent interest rate, but there would be some risk involved – he or she might not actually pay you your Tk. 1,000 on January 1,20x9.

Answer to the Question# 4 (b) (vi):

1/1/x5	8%/2	1/1/x6	1/1/x7	1/1/x8	1/1/x9
		400	PMT	PMT	PMT
					FV =1,000

We need to find the future value of the original Tk. 400 deposit:
This means that on January 1, 20x9, you need an additional sum of Tk. 493.87.

Tk. 10,000.00 – Tk. 506,13 = Tk. 493.87.

This will be an accumulated by making six equal payments that earn 8 percent compounded semiannually, or 4 percent each six months:

N = 6; I = 4%; PV = 0; FV = Tk. 493.87; PMT=?;
PMT=74.46.

Answer to the Question# 5 (a):

The general method for valuing a share of stock is to find the present value of all expected future dividends. The dividend growth model presented in the text is only valid (i) if dividends are expected to occur forever, that is, the stock provides dividends in perpetuity, and (ii) if a constant growth rate of dividends occurs forever. A violation of the first assumption might be a company that is expected to cease operations and dissolve itself some finite number of years from now. The stock of such a company would be valued by applying the general method of valuation explained in this chapter. A violation of the second assumption might be a start-up firm that isn't currently paying any dividends but is expected to eventually start making dividend payments some number of years from now. This stock would also be valued by the general dividend valuation method explained in this chapter.

Answer to the Question# 5 (b) (i):

Projected net income Tk. 2,00,000

Less projected capital investments	(800,000)
Available residual	Tk. 1,200,000
Shares outstanding	200,000

DPS = Tk. 1,200,000/200,000 shares = Tk. 6 = D_i.

Answer to the Question# 5 (b) (ii):

EPS = Tk. 2,00,000/200,000 shares = Tk. 10.

Payout ratio = DPS/EPS = Tk. 6/Tk. 10 = 60%, or
Total dividends / NI = Tk. 1,200,000/Tk.2,000,000 = 60%

$$P_0 = \frac{\hat{D}_1}{k_s = g} = \frac{Tk.6}{0.14 - 0.05} = \frac{Tk.6}{0.09} = Tk.66.67$$

Answer to the Question# 5 (b) (iii):

Currently, Under the former circumstances, $\hat{D}_1$ would be based on a 20 percent payout on Tk. 10 EPS, or Tk. 2. With Ks = 14% and g = 12%, we solve for P0:

$$P_0 = \frac{\hat{D}_1}{k_s = g} = \frac{Tk.6}{0.14 - 0.12} = \frac{Tk.2}{0.02} = Tk.100$$

Although CMC has suffered a severe setback, its existing assets will continue to provide a good income stream. More of these earnings should now be passed on to the shareholders, as the slowed internal growth has reduced the need for funds. However, the net result is a 33 percent decrease in the value of the shares.

Answer to the Question# 5 (b) (iv):

If the payout ratio were continued at 20 percent, even after internal investment opportunities had declined, the price of the stock would drop to Tk. 2/ (0.14 - 0.06) = Tk. 25 rather than to Tk.66.67. Thus, an increase in the dividend payout is consistent with maximizing shareholder wealth.

Because of the downward-sloping IOS curve, the greater the firm's level of investment, the lower the average ROE. Thus, the more money CMC retains and invests, the lower its average ROE will be. We can determine the average ROE under different conditions as follows:

Old situation (with founder active and 20 percent payout):

g = (1.0 - Payout ratio) (Average ROE)

12% = (1.0 - 0.2) (Average ROE)

Average ROE =12%/0.8 = 15% >ks= 14%.

Note that the average ROE is 15 percent, whereas the marginal ROE is 1 to 14 percent.

New situation (with founder retired and a 60 percent payout):

g = 6% = (1.0 – 0.6) (ROE)

ROE = 6% / 0.4 = 15% > k5 = 14%

This suggests that the new payout is appropriate and that the firm is taking on investments down to the point at which marginal returns are equal to the cost of capital.

---The End---